

Family name: Berbachir First Name: Mariam Group: lecture

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- Scientific calculators are not permitted during the exam.
- Only basic calculators, can be used.
- Answers written with a red pen or pencil will not be marked.
- Do not divide the page into two columns.

Exercise 1 (1 + 1.5 + 1 point).

Solve in $\mathbb{R}$.

- $|x-1| \leq x^2 + x + 2$.
- $||x-1| - 2| \leq 3$.
- Show that for all $x, y \in [0, 1]$, we have $x^2 + y^2 - xy \leq 1$.

1) if $x \geq 1 \Rightarrow |x-1| = x-1$. then:

$$|x-1| \leq x^2 + x + 2 \Leftrightarrow x-1 \leq x^2 + x + 2$$

$$\Leftrightarrow x^2 + 3 \geq 0 \text{ true } \forall x \geq 1.$$

if $x < 1 \Rightarrow |x-1| = -x+1$. then

$$|x-1| \leq x^2 + x + 2 \Leftrightarrow -x+1 \leq x^2 + x + 2$$

$$\Leftrightarrow x^2 + 2x + 1 \geq 0 \Leftrightarrow (x+1)^2 \geq 0 \text{ True, } \forall x < 1$$

Thus $S = \mathbb{R}$ is the set of solutions.

2) if $x \geq 1 \Rightarrow |x-1| = x-1$. then

$$||x-1| - 2| \leq 3 \Leftrightarrow |x-1-2| \leq 3 \Leftrightarrow$$

$$|x-3| \leq 3 \Leftrightarrow -3 \leq x-3 \leq 3$$

$$\Leftrightarrow 0 \leq x \leq 6 \text{ so solutions are:}$$

$$1 \leq x \leq 6, \text{ for } x \geq 1.$$

if $x < 1 \Rightarrow |x-1| = -x+1$, then:

$$||x-1|-2| \leq 3 \Leftrightarrow |-x+1-2| \leq 3$$

$$\Leftrightarrow |-x-1| \leq 3 \Leftrightarrow -3 \leq x+1 \leq 3$$

$$\Leftrightarrow -4 \leq x \leq 2, \text{ but } x < 1$$

then $-4 \leq x < 1$ is the set of solutions!

$$S = [-4, 1] \cup [1, 6] = [-4, 6].$$

$$3) \forall x, y \in [0, 1] \Rightarrow (1-x)(y-1) \leq 0$$

$$\Leftrightarrow y-1-xy+x \leq 0$$

$$\Leftrightarrow x+y-xy \leq 1, \text{ but } \forall x, y \in [0, 1]$$

we have $x^2 \leq x$, and $y^2 \leq y$, so:

$$x^2+y^2 \leq x+y \Rightarrow$$

$$x^2+y^2-xy \leq x+y-xy \leq 1.$$

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Exercise 2 (1 + 1 + 1 + 1 point).

Solve in $\mathbb{R}$.

- $\lfloor [x] \rfloor = x - 1$.
- $[x]^2 = [x]$.
- Prove that if $\forall n \in \mathbb{N} : [nx] = [ny]$, then $x = y$.
- Prove that: $\forall n \in \mathbb{N}, [(\sqrt{n} + \sqrt{n+1})^2] = 4n + 1$.

1. we know that $\lfloor [x] \rfloor = [x]$, thus the equation takes the form:

$$[x] = x - 1$$

but there exists a unique $n \in \mathbb{Z}$ such that $x = n + r$, with $r \in [0, 1[$.

then:

$$n = n + r - 1 \Leftrightarrow 0 = r - 1 \Leftrightarrow r = 1$$

impossible.

the set empty is the solution set.

$$2. [x]^2 = [x] \Leftrightarrow n^2 = n \Leftrightarrow n(n-1) = 0$$

$$\Leftrightarrow n = 0 \vee n = 1.$$

1. we deduce that $S = [0, 2[$ is the set solutions.

... one has:

$$[nx] \leq nx < [nx] + 1$$

$$[ny] \leq ny < [ny] + 1$$

$$\Rightarrow [nx] - [ny] - 1 < nx - ny < [nx] - [ny] + 1$$

by: $[nx] = [ny]$, so:

$$\forall n \in \mathbb{N} : -1 < nx - ny < 1$$

$\Leftrightarrow$

$$\forall n \in \mathbb{N} : -\frac{1}{n} < x - y < \frac{1}{n}$$

when n goes to $+\infty$, we get $x = y$.

...)

$$\forall n \in \mathbb{N} : [(\sqrt{n} + \sqrt{n+1})^2] = 4n + 1$$

$$\Leftrightarrow 4n + 1 \leq (\sqrt{n} + \sqrt{n+1})^2 < 4n + 2$$

$$\Leftrightarrow 4n + 1 - 1 \leq n + n + 2\sqrt{n(n+1)} < 4n + 2 - 1$$

$$\Leftrightarrow 2n + 1 - 1 \leq 2\sqrt{n(n+1)} < 2n + 2 - 1$$

$$\Leftrightarrow \cancel{4n^2} / \cancel{4n^2} + \cancel{4n} + 1 \leq \cancel{4n^2} + \cancel{4n} + 1$$

$$\Leftrightarrow 2n \leq 2\sqrt{n(n+1)} < 2n + 1$$

$$\Leftrightarrow 4n^2 \leq 4n^2 + 4n < 4n^2 + 4n + 1$$

since we have equivalence, this finishes the solution.

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Exercise 3 (1.5+2 point).

Let A be nonempty and upper bounded set of $\mathbb{R}$.

- 1- Prove that if $\sup A = a$, and $a \notin A$. Then a is an accumulation point of A .
- 2- Let $\Omega = \{\frac{n-1}{2n+1}, n \in \mathbb{N}\}$. Show that it's bounded and find its extremums. Use characterization when applicable.

1) $\sup A = a \Leftrightarrow \forall \varepsilon > 0, \exists x_\varepsilon \in A: a - \varepsilon < x_\varepsilon \leq a$.

since $a \notin A \Rightarrow$

$\sup A = a \Leftrightarrow \forall \varepsilon > 0, \exists x_\varepsilon \in A: a - \varepsilon < x_\varepsilon < a$.

a accumulation point of $A \Leftrightarrow$

$\forall \varepsilon > 0, \exists a - \varepsilon, a + \varepsilon[\setminus \{a\} \cap A \neq \emptyset$.

1.5 From the characterization of $\sup A$ we deduce that: $\exists a - \varepsilon, a + \varepsilon[\setminus \{a\} \cap A \neq \emptyset$, at least $x_\varepsilon \in A$ is a common element between the given general neighborhood $\exists a - \varepsilon, a + \varepsilon[$, and A .

$$2) A = \left\{ \frac{n-1}{2n+1}, n \in \mathbb{N} \right\} = \left\{ \frac{1}{2} - \frac{3}{4n+2}, n \in \mathbb{N} \right\}$$

$$n \geq 1 \Leftrightarrow 4n \geq 4 \Leftrightarrow 4n+2 \geq 6$$

$$\Leftrightarrow 0 < \frac{1}{4n+2} \leq \frac{1}{6} \Leftrightarrow 0 < \frac{3}{4n+2} \leq \frac{3}{6}$$

$$\Leftrightarrow -\frac{1}{2} < -\frac{3}{4n+2} < 0$$

$$\Leftrightarrow 0 \leq \frac{1}{2} - \frac{3}{4n+2} < \frac{1}{2}$$

From where, we deduce that A is bounded.

$$0,5 \quad \min A = \inf A = \frac{1}{2} - \frac{3}{4+2} = 0. \quad (n=1)$$

and $\frac{1}{2}$ is an upper bound, let us show that it is the least one.

$$\sup A = \frac{1}{2}.$$

$$1) \forall n \geq 1, \frac{n-1}{2n+1} < \frac{1}{2}.$$

$$2) \forall \varepsilon > 0, \exists n_0 \in \mathbb{N}: \forall n > n_0, \frac{1}{2} - \varepsilon < x_n < \frac{1}{2}.$$

Indeed: for given $\varepsilon > 0$, does n_0 exist?

$$\frac{1}{2} - \varepsilon < \frac{1}{2} - \frac{3}{4n+2}$$

$$\Leftrightarrow -\varepsilon < -\frac{3}{4n+2} \Leftrightarrow \varepsilon > \frac{3}{4n+2}$$

$$\Leftrightarrow 4n > \frac{3}{\varepsilon} - 2 \Leftrightarrow n > \frac{1}{4} \left(\frac{3}{\varepsilon} - 2 \right)$$

It is sufficient to take:

$$n_0 = \left[\frac{1}{4} \left(\frac{3}{\varepsilon} - 2 \right) \right] + 1.$$

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Exercise 4 (1.5 + 1 + 1.5 point).

1- Calculate the moduli and the Argument of $z_1 = 1 + i(1 + \sqrt{2})$.

2- Check the following equality :

$$\left(\cos\left(\frac{\pi}{7}\right) + i \sin\left(\frac{\pi}{7}\right)\right) \left(\frac{1 - i\sqrt{3}}{2}\right) (1 + i) = \sqrt{2} \left(\cos\left(\frac{5\pi}{84}\right) + i \sin\left(\frac{5\pi}{84}\right)\right)$$

3- Calculate the moduli and the Argument of u, v , where: $u = \frac{\sqrt{6} - i\sqrt{2}}{2}$ and $v = 1 - i$.

Deduce the moduli and the Argument of $\frac{u}{v}$.

1- we can remark that $z_1 = 1 + i + i\sqrt{2}$

let consider $h_1 = 1 + i$, and $h_2 = i\sqrt{2}$

then $|h_1| = \sqrt{2}$, $|h_2| = \sqrt{2}$, and

$\text{Arg } h_1 = \frac{\pi}{4}$, $\text{Arg } h_2 = \frac{\pi}{2}$. Thus:

$$z_1 = \sqrt{2} (e^{i\frac{\pi}{4}} + e^{i\frac{\pi}{2}}) = \sqrt{2} e^{i\frac{3\pi}{8}} (e^{i\frac{\pi}{8}} + e^{i\frac{\pi}{8}})$$

$$= 2\sqrt{2} \cos \frac{\pi}{8} \cdot e^{i\frac{3\pi}{8}} \quad \text{so } |z_1| = 2\sqrt{2} \cos \frac{\pi}{8}$$

and $\text{Arg } z_1 = \frac{3\pi}{8}$

$$|z_1| = 2\sqrt{2} \cos \frac{\pi}{8}$$

2- one has:

$$* \cos \frac{\pi}{7} + i \sin \frac{\pi}{7} = e^{i\frac{\pi}{7}}$$

$$* \frac{1 - i\sqrt{3}}{2} = e^{-i\frac{\pi}{3}}$$

$$* 1 + i = \sqrt{2} e^{i\frac{\pi}{4}}$$

then:

$$\textcircled{1} \quad \left(\cos \frac{\pi}{7} + i \sin \frac{\pi}{7} \right) \left(\frac{1 - i\sqrt{5}}{2} \right) (1+i) = e^{i\frac{\pi}{7}} \times e^{-i\frac{\pi}{3}} \times \sqrt{2} e^{i\frac{\pi}{4}} \\ = \sqrt{2} e^{i\frac{5\pi}{84}} \quad \text{True.}$$

$$3 - |u| = \sqrt{2}, \quad |v| = \sqrt{2} \rightarrow 0,5$$

$$\text{Arg } u = -\frac{\pi}{6}, \quad \text{Arg } v = -\frac{\pi}{4} \rightarrow 0,5$$

$$\text{Arg } \frac{u}{v} = -\frac{\pi}{4}, \quad \left| \frac{v}{v} \right| = 1 \rightarrow 0,5$$

$$\text{Arg } \frac{u}{v} = -\frac{\pi}{6} + \frac{\pi}{4} = \frac{-2\pi + 3\pi}{12} = \frac{\pi}{12}$$